

Spring DATA JPA

- ♦ CRUD Operations called Insert, Update, Retrieve and Delete Operations are common across all DAOs. The only difference is domain types you are interacting with.
- ♦ Spring Data internally provides the implementation for all such common methods. So developer is not responsible for writing implementation for these common methods.

Ex:

```
public interface CustomerDAO extends JpaRepository<Customer, Integer>{  
    //Empty  
}
```

- ♦ JpaRepository is parameterized and takes two parameters
 - Entity class
 - Primary Key type
- ♦ JpaRepository extends CrudRepository.
- ♦ Both CrudRepository and JpaRepository has 15+ methods for performing common persistence operations such as saving, deleting, finding

Interface CrudRepository<T, ID>

- 1) S save(S entity)
- 2) void deleteById(ID id)
- 3) void delete(T entity)
- 4) void deleteAll(Iterable<? extends T> entities)
- 5) void deleteAll()
- 6) Optional<T> findById(ID id)
- 7) long count()
- 8) boolean existsById(ID id)

Interface JpaRepository<T, ID>

- 1) S saveAndFlush(S entity)
- 2) List<S> saveAll(Iterable<S> entities)
- 3) List<T> findAll()
- 4) List<T> findAllById(Iterable<ID> ids)
- 5) List<T> findAll(Sort sort)
- 6) void flush()
- 7) T getOne(ID id)



demo1-data-jpa: Files required

1. MyBootApplication.java	Updated in Demo1
2. Customer.java	Same as Previous Demos
3. CustomerDAO.java	Updated in Demo1
4. application.properties	Same as Previous Demos
5. pom.xml	Same as Previous Demos

1. MyBootApplication.java

```
package com.jlcindia.springboot;
```

```
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.boot.CommandLineRunner;  
import org.springframework.boot.SpringApplication;  
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @company : Java Learning Center
```

```
*/
```

```
@SpringBootApplication
```

```
public class MyBootApplication implements CommandLineRunner {
```

```
@Autowired
```

```
CustomerDAO custDAO;
```

```
@Override
```

```
public void run(String... args) throws Exception {
```

```
System.out.println(custDAO.getClass());
```

```
// 1.save() method - inserting
```

```
Customer cust = new Customer("bbb", "ccc@jlc", 222, "Blore");
```

```
custDAO.save(cust);
```

```
System.out.println(cust.getCid());
```

```
//2. saveAndFlush() - A) inserts if P.K not given
```

```
Customer cust = new Customer("ccc", "ccc@jlc", 333, "Blore");
```

```
custDAO.saveAndFlush(cust);
```

```
//3. saveAndFlush() - B) updates if P.K given and present
```

```
Customer cust = new Customer(11,"bbb", "bbb@jlc", 222, "Blore");
```

```
custDAO.saveAndFlush(cust);
```



//4. **saveAndFlush()** -C) Inserts if P.K given and not present

```
Customer cust = new Customer(99,"sri", "sri@jlc",1234567, "Hyd");  
custDAO.saveAndFlush(cust);
```

```
System.out.println("Done!!!");
```

```
}
```

```
public static void main(String[] args) {  
    SpringApplication.run(MyBootApplication.class, args);  
}
```

```
}
```

2. Customer.java

3. CustomerDAO.java

```
package com.jlcindia.springboot;
```

```
import org.springframework.data.jpa.repository.JpaRepository;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @company : Java Learning Center
```

```
**/
```

```
public interface CustomerDAO extends JpaRepository<Customer, Integer>{
```

```
}
```

4. application.properties

#Database Properties

```
spring.datasource.driver-class-name=com.mysql.jdbc.Driver
```

```
spring.datasource.url=jdbc:mysql://localhost:3306/myjlcdb5
```

```
spring.datasource.username=root
```

```
spring.datasource.password=srinivas
```

#JPA Properties

```
spring.jpa.hibernate.ddl-auto=update
```

```
spring.jpa.show-sql=true
```

#Log Properties

```
logging.level.root=INFO
```

```
logging.pattern.console=%-5level %logger{36} - %m %n
```



5. pom.xml

<project...>

```
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo1-data-jpa</artifactId>
<version>1.0</version>
<name>demo1-data-jpa</name>
<description>This is Spring Data JPA Example</description>
<url>https://www.jlcindia.com</url>
```

```
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>
```

<dependencies>

```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
```

```
<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>
```

```
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```